

- (iii) Explain why two gamma-ray photons are produced, rather than just one.

.....
 [1]

- (iv) Show that the kinetic energy of the electron before the collision is $1.1 \times 10^{-15} \text{ J}$.

[1]

- (v) Use the information in (d)(iv) to determine, to three significant figures, the wavelength associated with the gamma radiation emitted in the collision.

wavelength = m [3]

[Total: 13]

